Clap Based Light Intensity Control Using Arduino

Report Submitted to
National Institute of Technology Manipur
For the award of the degree

of

Bachelor of Technology

In Electronics & Communication Engineering

By

Harshajeet Rabha (21105029)

Shinguingam Phaomei (21105037)

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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

NATIONAL INSTITUTE OF TECHNOLOGY, MANIPUR

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Department of Electronics and Communication Engineering
राष्ट्रीय प्रौद्योगिकी संस्थान मणिपुर
National Institute of Technology Manipur
(An autonomous Institute under Ministry of Education, Govt. of India)

Ref no.:-……. Dated:-…….

SUPERVISOR CERTIFICATE

This is certified that the dissertation report entitled, "**Clap Based Light intensity control using Arduino**" submitted by ***Harshajeet Rabha (21105029), Shinguingam Phaomei (21105037) and R D Thepii Anthony Vemai (21105005)*** submitted to National Institute of Technology Manipur, India is a record of Bonafide Project Work-3 carried out by them under my supervision and is worthy of consideration for fulfillment of 8th semester Project work-3 of **Bachelor of Technology in Electronics and Communication Engineering** of the Institute.

Date: -……..

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CERTIFICATE

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STATEMENT

I do here by declare that the matter embodied in this thesis entitled **"Clap Based Light intensity control using Arduino**" is the result of investigation carried out by me under the supervisor of **Dr. Ashish Ranjan** in the department of Electronics and Communication Engineering National Institute of Technology-Manipur. India.

In keeping with the general practice of reporting scientific observation, due acknowledgements have been made wherever the work described is based on the findings of other investigators.

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DECLARATION

We the undersigned, certify that

a. The work contained in this report is original and has been done by us under the guidance of ***Dr. Ashish Ranjan, Associate Professor,*** *Department of Electronics and Communication Engineering*.

b. The work has not been submitted to any other Institute for any degree or diploma.

c. We have followed the guidelines provided by the Institute in preparing the report.

d. We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

e. Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.

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Abstract

Developing projects to facilitate the controlling of electronic devices is a critical concern nowadays. The project's goal is to develop "light intensity clappers" in which light systems can be managed by hand claps. The main components of the project are based on Arduino.

The objective of the project "Clap-Based Light Intensity Control Using Arduino" is to create a highly efficient application that makes it easy to control the brightness of a light source using sound input, particularly by clapping, which makes it more energy-efficient.

With this system, sounds sensors can pick up the sound of a clap and convert them to electrical signals, where it will then be processed by Arduino UNO. Note that after each clap, the Arduino will command the light's driver to dim or brighten the light.

The secondary objectives for this project are as follows: to be more cost effective and adaptable through maintenance and modification further increasing its workability as a home automation system; beneficial for persons with movement disabilities and for environments where hand operated switches are not preferred.

Home automation, smart lighting systems and other energy conserving mechanisms are some of the most noteworthy uses for this project. This project can serve as an input towards more advanced development of control system based on sound

CHAPTER 1

INTRODUCTION

1.1 Background

The invention, “Clap-Based Light Intensity Control Using Arduino” represents a practical idea that is fully automated which can be applied in the automation of lighting systems in different places. Its main objective is to improve convenience and energy saving by allowing users to command the setting level of brightness of a light source actively via sound, in this case, through claps.

This system primarily relies on a sound sensor module that is capable of recognizing clap sounds with high accuracy. The sound sensor will inform an Arduino UNO microcontroller about a clap when it’s detected. The sound sensor monitors the magnitude of the in coming sound or noise signal with respect to time. The Arduino in this case is preset to certain clapping signals, executes specific activities by altering the light level of a lamp attached to Arduino through PWM. This gives the possibility of altering the degree of brightness of the light with marginal adjustments which smoothens the operation and further enhances comfort of the users.

The project has an innovative design which is cheap and easy to scale up, characteristics that make it ideal for home systems. This design offers a simplistic method for people with mobility difficulties to adjust lighting without the need to struggle with switches or dials. In addition, these types of operations are advantageous in need of sanitization, such as hospitals or labs.

Beyond simple home lighting, the project has numerous applications. They can be added to smart home systems which improves user experience and helps control household devices in a more intuitive way. This will also help with electricity cost and consumption making the device more environmentally friendly.

In conclusion, the project “Clap-Based Light Intensity Control Using Arduino” provides an intelligent solution to lighting but it can also be used as an introduction for advanced work in automation and sound activated controls technologies.

1.2 Objective

- Developing projects to facilitate the controlling of electronic devices is a critical concern nowadays. The project's goal is to develop "light intensity clappers" in which light systems can be managed by hand claps. The main components of the project are based on Arduino.

- The objective of the project "Clap-Based Light Intensity Control Using Arduino" is to create a highly efficient application that makes it easy to control the brightness of a light source using sound input, particularly by clapping, which makes it more energy-efficient.

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- Home automation, smart lighting systems and other energy conserving mechanisms are some of the most noteworthy uses for this project. This project can serve as an input towards more advanced development of control system based on sound.

1.3 Scope

This thesis focuses on the design, development, implementation and control of dimming or brightening a light with a sound activated controlled using Arduino Uno. The specific areas that are been considered for this project include:

1.3.1 Educational values:

- Engages with and resolves challenges related to Arduino programming, sensor interfacing, and electronics, providing a solid understanding of these foundational areas.
- It explores complexities in sound signal processing and the application of PWM (Pulse Width Modulation) techniques to effectively manage light control systems, enhancing both functionality and creativity in electronic projects.

1.3.2 Core Components Involved:

- Arduino Uno: A popular platform for creating and controlling a wide variety of electronic projects.
- Sound Sensor: A sensor that measures the intensity of sounds in its surroundings and can control electronic circuits accordingly.
- LED: A low-power LED commonly used to signal the outcome or state of electronics in prototype circuits.

1.3.3 Functional Scope:

- The system is designed to detect one or multiple claps by analyzing data from sound sensors. Once a clap is identified, it can trigger various lighting functions, such as turning the lights ON or OFF.
- The system is capable of adjusting the brightness levels in response to consecutive claps, allowing for gradual dimming or brightening with each clap detected. This interactive feature not only enhances user experience but also provides a seamless way to control lighting environments.

1.3.4. Applications:

- Home Automation Systems: These systems utilize sound-based controls to enable seamless management of various household functions, enhancing convenience and efficiency.
- Hands-Free Lighting Solutions: Designed to improve accessibility and hygiene, these systems allow users to operate lights without physical contact, making them ideal for individuals with mobility challenges or when hands are occupied or dirty.
- Ambient Lighting Systems: Perfect for smart rooms or theaters, these systems create dynamic lighting environments that can adjust based on mood, activity, or time of day, enriching the overall experience.
- Energy-Saving Systems: These intelligent systems detect periods of inactivity and automatically turn off lights when no sound such as a clap or voice command is detected, contributing to energy conservation and reducing electricity costs.

1.3.5. Future Extensions:

- Enable people to control the system effortlessly with their voices instead of using their hands. Voice control makes it easy for users to interact with the system without any challenges.
- Changing the brightness of the lamp is possible simply by clapping in a predetermined way. Combinations of claps will allow the system to adjust the lighting in response to each user's desired atmosphere. This distinctive way of managing the system enables people to interfere easily and significantly enhances their experience using the service.

1.4. Motivation behind the project:

Currently, saving energy, making things easier for users, and using automation are the primary goals for smart homes. Proper lighting makes any space better, and improving how we use it can considerably help with energy usage and what users experience. Conventional lights have just two settings: on or off, so there is little chance to control the light level and they consume power unnecessarily much of the time. In addition, many times users must use a physical switch, which may not always be possible for everyone due to physical restrictions or age.

To overcome these issues, experts and enthusiasts now use microcontrollers to manage how lights are controlled within a home. An example of this is creating a way to control lighting brightness through clapping. Clapping can instantly change the brightness of the light, allowing for a comfortable and economical way to adjust the lighting. Because of an increased need, the new system should be able to adapt to changes in its environment or to user inputs even without touch buttons. Once the loud clap is sensed by the microphone module, the Arduino acts on it and changes the light's intensity or state. Since it has precise control over light, this design is at once user-friendly and saves energy because it adjusts to what is required or happens in the room.

The project represents many technical problems from an engineering view. The Arduino first identifies the clap (as distinct from other sounds), processes the sound data, and finally outputs an action to change the brightness of the light. It is also necessary to ensure that the system works smoothly, keeps false alarms down, and fits properly with the existing lighting system. It is also important to note that such a project benefits people educationally. It helps to understand embedded systems, signal processing, various electronic parts and connecting hardware and software to deal with real-life situations.

Students and hobbyists who work on this type of project learn about circuit building, programming using Arduino, and common debugging solutions, turning it into a complete learning experience. In essence, the reason for this project is to create smarter, easier, and more power-saving ways to control lighting. It creates a connection between how things are actually used and the theoretical approach in electronics and programming. With this project, we can see how easy actions like a clap can help us control technology more naturally and help it become more available to all.

1.5. Overview of research method

1.5.1 Problem Identification:

Creating a system that enables people to increase the brightness of the light simply by talking is extremely beneficial. Consequently, this solution offers ease of use for everyone.

1.5.2. Literature Review:

Exploring existing studies and contemporary technologies allows understanding the limitations and advantages involved in building the system.

1.5.3. System Design:

- Hardware Design: All the components of the circuit must be properly connected to get the system working.
- Circuit Design: Designing a schematic diagram to visualize how the sound sensor, LED and Arduino communicate with each other.

One clap corresponds to a quarter of the brightness being displayed by the LED.

Heard two claps, the level of light intensity is raised by half.

The intensity gets to 75% after making three claps successively.

The light will be bright after detecting four consecutive claps.

Five Times clap Switches Off the Light

1.5.4. Prototype Development:

At this point, connect all components, load the Arduino code and see if the project is functioning as intended.

1.5.5. Testing and Calibration:

Testing the accuracy of clap detection, evaluating the speed at which the light responds, monitoring the system's ability to reduce brightness and assessing how reliable the overall system is. The system undergoes changes in the software and hardware as a result of the testing process.

1.5.6. Data Collection and Analysis:

Assessing the system's performance based on its reactions to claps and the feedback received from individuals.

1.5.7. Result Evaluation:

Verifying that the system is functioning correctly by assessing the adjustments in lighting whenever people clap

CHAPTER-2

LITERATURE SURVEY

2.1. Papers

2.1.1 Geethika et al. [1] proposed a new system of automatic control of light intensity using Arduino microcontrollers and light-dependent resistors (LDRs). It was primarily intended to promote energy efficiency by avoiding unnecessary power consumption, mitigating eye strain caused by unsuitable lighting conditions, and maximizing the utilization of available daylight. The mechanism is based on continuous measurement of ambient light levels by the LDRs, which in turn instruct the Arduino to regulate artificial light levels accordingly. Although the system shows great potential for improving lighting efficiency and user comfort, it is not without several drawbacks. The installation is quite complicated with a necessary judicious integration of sensors, wiring, and control logic. Also, the total cost can be expensive, especially when magnified to larger environments or high-quality components are utilized. Also, the system depends on natural light which means the operation relies on environmental conditions such as time of day or weather, which will have a high chance of operation failure.

2.1.2 N. Khera et al. [2] developed a microcontroller-based system which uses PWM signals and PIR sensors to adjust LED brightness based on motion, saving energy. One of the advantages of this system is its ability to be remote controlled, facilitated by the interfacing of LabVIEW software with the internet in such a way that users are able to access and control light conditions from other places in real time.

The system relies on stable internet connectivity for remote control that may be a drawback in areas of poor or unreliable network infrastructure, resulting in intermissions or delays in control operations. Additionally, the use of PIR sensors for movement detection, as excellent as they are, may fail to detect stationary or minimal human presence, resulting in unwanted dimming or turning off of lights.The system is not expandable with ease, thus it is less ideal for application in extensive or complicated sites such as multi-story buildings

or factories unless it is modified or renovated.

2.1.3 Kaku et al. [3] described the construction and operation of an intelligent lighting system in a real office environment by using Lighting fixtures, illuminance sensors, dimmers, and a control PC. The goal of this project is to enhance energy efficiency and create a comfortable lighting environment for office workers by automatically adapting the brightness as per the ambient light environment and workspace layout.

The system performes well in formal office environments where the layout is set and activities go in a planned sequence. Within these controlled environments, the lighting needs are relatively standard, and it is simpler for the control system to provide the best lighting. While effective in the controlled environment, though, the system is less adaptable. Implementation of this lighting system in a variety of other types of settings residential, dynamic offices, or factory floors would involve major hardware and software adjustments. Room configurations' adaptation, varying lighting needs, and easier or independent user control needs would demand more scalable and flexible configuration. Therefore, the existing system fits best in fixed office setups and might not as well or even be feasible in more unstable or unstable environments without considerable upgradation or overhaul.

2.1.4 Torres-Maldonado, Sergio Yahel, et al. [4] discussed the designing and implementing a voice assistant within IoT-based home automation systems. The main aim of their system is to enhance the overall convenience, security, and energy efficiency in intelligent homes by enabling users to manage multiple household devices like lights, alarms, and environmental sensors via voice commands as well as an easy-to-use web interface. This integration offers a user-friendly and hands-free solution for coping with everyday chores, most especially useful to users who have mobility issues or those who would just like things automated for comfort and convenience.

The system shows capable performance in controlled environments and with extremely clear voice commands. It also has a few weaknesses, most notably in regard to voice recognition accuracy. One of the primary limitations is its relatively high error rate, particularly when interpreting voice inputs from users

with different tones, accents, or speech patterns. This can result in misinterpretation of commands or inability to recognize them at all, decreasing the reliability of the system in diverse linguistic environments. In addition, in the case of users with the same kind of accents giving commands, the system could end up misinterpreting a single command as applicable to more than one device. This can lead to unintended activation or deactivation of a number of home appliances simultaneously, causing inconvenience or even possible safety issues. Making the system stronger and universal would require enhanced natural language processing, accent recognition, and contextual interpretation of commands. Without such improvement, the system can become difficult to execute consistently with various user populations and real-world environments.

2.1.5 Rath, Deepak Kumar et al. [5] designed and developed an automatic lighting control system with the help of the Arduino UNO microcontroller and light-dependent resistors (LDRs) in order to control lighting automatically as a function of ambient level of light in real time. The system is designed to measure the amount of luminous energy present within the immediate surroundings at any given moment and adjust the amount of the affixed lighting system accordingly. This system operates to conserve energy through minimizing power redundancy during the daytime or under normal natural light, all while providing sufficient lighting under the low light periods.

Yet, insofar as they succeed in constant conditions, the system's functioning may be impacted by abrupt alterations of natural lighting, like cloud or moving sun, which create too many recurrent readjustments. In addition, the system has no high-level features such as user access control, scheduling, or interface to other intelligent home technologies, that might confine its functionality for more complex or customized implementations. In spite of these constraints, the project is a hands-on application of how one can apply basic electronics and microcontroller programming to build effective and responsive lighting systems.

2.1.6 Siregar et al. [6] created a remote light control system based on the Arduino Uno microcontroller. The system enables users to switch lights ON or OFF remotely through SMS commands. It saves time and effort by allowing control

based on location or time as long as there is cellular network coverage, the system remains accessible. This eliminates the need for physical presence to manage lighting. It is especially useful for improving energy management and security.

The design is simple, cost-effective, and suitable for basic home automation

2.1.7 Lestari et al. [7] created a smart home system for light control system applications using the Arduino Uno microcontroller via mobile-based Wi-Fi media. With this application, the user can control the home lights by turning them off or on remotely through a mobile device, thereby improving the efficiency of electricity usage. The system is intended to be accessible and user-friendly and can be controlled in real-time with no complicated wiring or manual switches. It facilitates improved energy conservation through the function of ensuring that lights are utilized only when necessary. Wi-Fi connectivity, being utilized, also facilitates integration with other intelligent devices, increasing the scalability of the system for wider home automation. Also, the mobile app has a very easy interface that can be accessed by individuals of any age. The system is quite effective for individuals who often tend to forget turning off lights when exiting home. It also helps promote home security through the ability to control lights even when a user is out, such that it gives an impression that there are people at home.

2.1.8 Darma, Mirza Surya et al. [8] created a sound sensor-based automation of light with an Arduino microcontroller where FC-04 sound sensor detects specified sounds such as claps or voice commands and gives input to Arduino to control lights. A relay that is utilized as an electronic switch is controlled by the Arduino to turn the lights on or off accordingly. This free-hand system is more convenient for users and is particularly handy for people with mobility impairment, offering a convenient alternative to manual switches. It also supports energy saving since the lights are used only when needed. It is easy to install and cheap, and this makes it suitable for small DIY smart home projects. However, the system sometimes responds to unwanted background noise, and caution must be exercised while calibrating the sensitivity of the sensor. Despite this limitation, the system works without an internet connection and works best within controlled indoor environments like bedrooms or offices.

2.1.9 Mumtaz, Zain, et al. [9] proposed an Arduino-based automated system that is intended to regulate streetlights wisely by employing solar light detection and object sensing technology. Under the initial configuration, the system automatically turns on the streetlights to a DIM state at night and brightens them to a HIGH state when an object, e.g., vehicle or pedestrian, is sensed. In day time, the streetlights are kept OFF, as there is natural light. This practice minimizes wasteful energy use and extend the life of lighting elements. For further optimization, the system was then revised to exclude the DIM mode completely during nighttime. In the revised version, the streetlights are OFF throughout the night unless an obstacle is sensed, in which case they turn directly to the HIGH state. This development not only saves more energy but also ensures that lighting is made available only as and when required, thus being extremely appropriate for smart city and energy conservation applications. The system presents a cost-saving, automatic, and eco-friendly option for controlling streetlights.

2.1.10 Wanda, Wanda, and Syahrul Mubaraketal. [10] created an automatic lamp control system using a sound sensor and an Arduino microcontroller. The smart lighting system offers voice control for lights, i.e., users can control lights by turning them on and off using the voice command of the room where lights are to be turned on or off. Because it recognizes voice commands, the system eliminates physical contact with switches or movement across different areas of the home. It is especially useful in multi-level homes, as users are able to command lights at the second or third floor without having to go up the stairs. This improves convenience, increases user comfort, and makes it easy for persons with mobility impairments. The design is particularly to facilitate easier daily lighting control and to make it more efficient. It also conserves time and energy through the provision of voice-controlled, centralized control. The system is well suited for modern smart homes seeking automation and minimization of manual intervention.

CHAPTER-3

REQUIREMENT ANALYSIS

3.1 Software Requirements:

The software Requirement we use in this project are:

3.1.1 ARDUINO IDE:

Arduino IDE tool (software) is an open-source environment where we can write code and upload it to UNO board. It runs on Windows Mac OS and Linux OS. The software environment required to program the Arduino by writing code and uploading it to The Arduino. It also provides results for analysis using both a serial monitor and a serial plotter.

A software environment is required to program the Arduino by writing the code and uploading it to the Arduino. It also provides results for analysis using both a serial monitor and a serial plotter. Arduino IDE used to write code Arduino IDE supports C and C++ languages using special code structuring rules. The Arduino IDE provides a software library from the Wiring Project, which provides many common input and output processes. User-written code requires only two basic functions, to initialize Sketch and the main program loop, which is compiled and linked to the program stub main () in a cyclic executable program executable with the GNU toolchain, is also included with the IDE distribution.

3.1.2 Components of the Arduino IDE

3.1.2.1 Sketchbook:

- Sketches: Programs written for Arduino are called sketches. Each sketch is saved with a. ino extension.
- Examples: The IDE includes a wide range of example sketches that demonstrate the use of various functions and libraries.

3.1.2.2 Code Editor:

- Syntax Highlighting: Different parts of the code are color-coded for better understandability.
- Auto-Completion: Suggests possible code completions and functions names as you type.
- Bracket Matching: Helps in matching opening and closing brackets.

3.1.2.3 Compiler:

- Verification: convert the written code (sketch) into a computer readable format. The “verify” button checks the code for errors without uploading.
- Message Window: Displays error messages and warnings during compilation.

3.1.2.4 Uploader:

- Upload Functionality: After verification, the compiled code can be uploaded to the connected Arduino board using the "Upload" button.

3.1.2.5 Serial Monitor and Plotter:

- Serial Monitor: used to send and receive serial data between the host computer and the Arduino board. It is useful for debugging as well as for real-time communication.
- Serial Plotter: The one which plots the serial data graphically is best for sensor data monitoring along time.

3.1.2.6 Library Manager:

- Standard Libraries: These are the libraries that come pre-installed with the IDE. They offer helpful tools for working with things like sensors, displays, and various communication protocols.
- Other Libraries: it also lets you easily browse, install, update, and manage additional third-party libraries right from within the IDE.

3.1.3 Programming with Arduino IDE

3.1.3.1 Sketch Basic Structure:

- Setup Function: The setup function is executed once whenever the board is powered on or reset. It is useful for initializing variables, setting pin modes, and commencing the usage of libraries, etc.
- Loop Function: This function stops after the loop has been entered once until it takes control continuously. It is used to control the Arduino board actively.

3.1.3.2 Common Functions:

- Digital I/O: Functions such as pinMode, digitalWrite and digitalRead are made to control and read from digital pins.
- Analog I/O: Functions like analogRead, analogWrite and analogReference are analog input/output function,

3.1.4 Getting Started

3.1.4.1 Install the Arduino IDE:

First, visit the respectable Arduino website and download the Arduino IDE. It's available for Windows, Mac, and Linux. Once downloaded then observe the setup instructions and set up it like any normal app.

3.1.4.2 Connect Your Arduino Board:

Take a USB cable and connect your Arduino board for your pc. When the board typically lights up to show that it is powered.

3.1.4.3 Set the Board and Port in the IDE:

Open the Arduino IDE. Click on Tools > Board and choose your unique Arduino version (for example, Arduino Uno). Then visit Tools > Port and select the right port (commonly classified as COM observed with the aid of quite a number). If you're unsure, unplug your board and plug it back in to peer which port seems.

3.1.4.4 Write or Load Your First Program:

If that is your first time, a incredible region to begin is with the built-in Blink example. It's a easy application that makes the onboard LED blink on and rancid each. You can locate it below File > Examples > Basics > Blink.

3.1.4.5 Verify and Upload Your Code:

Click the checkmark icon in the pinnacle-left corner of the IDE to test your code for any errors. If the whole lot appears excellent, hit the arrow icon to upload the code in your Arduino board.

3.2 Hardware Requirements:

Hardware Components:

1. Arduino UNO
2. Sound Sensor Module (LM393)
3. Jumper Wires
4. USB Cable
5. LED
6. 9V Battery

3.2.1 ARDUINO UNO:

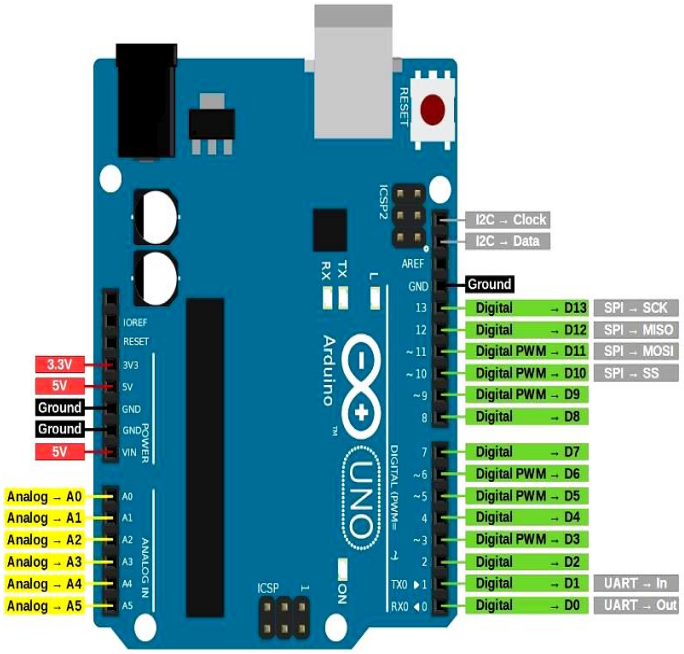


Figure 3.1: - Arduino UNO Diagram

The Arduino Uno is the primary microcontroller board used on this project. It receives input signals from the sound sensor and processes them according to the code uploaded to it. Based on the range of claps detection, it decides how much light intensity have to be applied. The Arduino is also responsible for generating PWM (Pulse Width Modulation) signals to manipulate the brightness of the LED or light bulb.

3.2.1.1 GENERAL PIN FUNCTIONS

3.2.1.1.1 Digital input/output Pins (0-13):

- These 14 pins can be used as both input or output the use of pinMode(), digitalRead(), and digitalWrite() inside the code.
- Pins 3, 5, 6, 9, 10, and are PWM output the usage of the analogWrite() function, useful for dimming LEDs or controlling automobiles.

3.2.1.1.2 Analog input Pins (A0-A5):

- These pins can read analog signals (0-5V) and convert them into digital values the use of the ADC (Analog-to-Digital Converter).
- Commonly used to study sensor values such as temperature, mild, or sound.

3.2.1.1.3 Power Pins:

- Vin: Input voltage to the Arduino while using an external power supply.
- 5V: Supplies regulated 5V from the board to sensors or modules.
- 3.3V: A lower voltage supply (use with low-voltage modules).
- GND: Common floor reference for the circuit.
- RESET: Used to manually reset the Arduino.

3.2.1.1.4 Communication Pins:

- TX (pin 1) and RX (pin 0): Used for serial conversation (UART). These are related to the USB-serial converter.
- A4 (SDA) and A5 (SCL) are used for I2C communication with multiple devices like sensors and many others.
- Pins 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK) are used for SPI (Serial Peripheral Interface) to talk with gadgets like SD cards or shift registers.

3.2.1.2 TECHNICAL SPECIFICATION OF ARDUINO UNO:

Table 3.1: - Technical specification of Arduino UNO

Feature	Specification	Bit Level
microcontroller	ATmega328P	8-bit AVR RISC architecture
Operating Voltage	5V	-
Digital I/O Pins	14	Controlled by 8-bit register
PWM Output Pins	6 (Pins:3,5,6,9,10,11)	Timer/Counter, 8-bit/16-bit
Analog Input Pins	6 (A0 to A5)	10-bit ADC resolution
Flash Memory	32 KB	8-bit wide access
SRAM	2 KB	8-bit word size
EEPROM	1 KB	8-bit non-volatile memory
Clock Speed	16 MHz	Drives 8-bit processor cycles

3.2.2 Sound Sensor Module (LM393):

The LM393 sound sensor is a module used to stumble on sound degrees in the surroundings. It doesn't degree how loud a legitimate is in decibels, but instead it detects if the sound crosses a sure level and then sends a virtual or analog signal. It's normally used in clap-based switches, voice activated structures, and basics sound detection projects.

3.2.2.1 Key functions of LM393

3.2.2.1.1 Digital and Analog output pins:

- Digital Output (D0): it sends a HIGH signal while the detected sound crosses the threshold value.
- Analog Output(A0): Continuously changes with sound intensity. Useful while you want to measure how loud the sound is, not just detect if its loud.

3.2.2.1.2 Microphone:

- Picks up sounds from the surroundings and sends it to the circuit for processing.

3.2.2.1.3 Adjustable Sensitivity:

- Includes a potentiometer that lets fine-tune how sensitive the sensor is to sound.

3.2.2.1.4 Power Supply:

- Operates on 3.3V to 5V, so it is compatible with maximum microcontrollers (Arduino, ESP32 etc.).

3.2.2.1.5 Onboard LED Indicator:

- Lights up when the detected sound level exceeds the threshold value.

3.2.2.1.6 Simple 4-Pin Interface:

- VCC – Power enter (3.3V–5V)
- GND – Ground
- A0 – Analog output
- D0 – Digital output

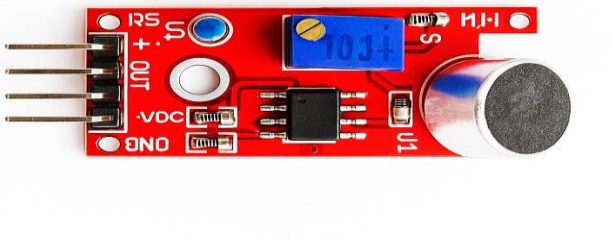


Figure 3.2: - LM393 Sound Sensor

3.2.3 JUMPER WIRES:

Jumper wires are simply wires that have connector pins at each end, which can be used to connect two points together without soldering. Jumper wires are commonly used with breadboards and other prototyping tools to make it easy to change circuits as needed.

3.2.3.1 Types of Jumper Wires:

1. Male-to-male jumper
2. Male-to-female jumper
3. Female-to-female jumper

There are two types of head shapes for jumper wires

1. Square head
2. Round head

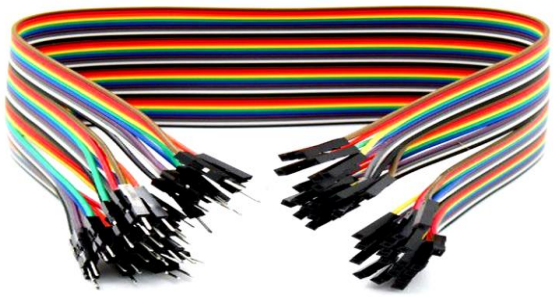


Figure 3.3:- Jumper Wires

3.2.4 USB CABLE:

A USB cable is one of the most important parts when you are working with an Arduino board like the Arduino Uno. It connects your Arduino to your computer and helps you do two main things like upload your code and powered the board.

Most of the time, you will use a USB Type-A to Type-B cable with Arduino Uno. For smaller boards like the Arduino Nano, a Mini USB or Micro USB cable is used depending on the version.

When you write code in the Arduino software on your computer, the USB cable lets you send that code to the Arduino board. It also supplies electric power to the board so that it can run the program. If you want to see messages or test values from the board (like sensor readings), the USB cable lets the board talk back to your computer using something called the Serial Monitor.

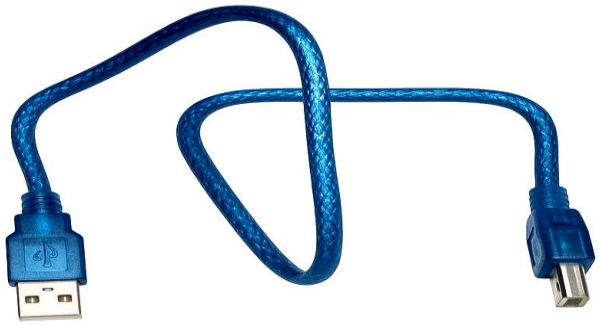


Figure 3.4: - USB Cable

3.2.5 LED:

An LED (Light Emitting Diode) is an electronics component that emits light while a small amount of electric current passes through it in the forward biased. It is a type of diode that convert electrical energy directly into light energy.

3.2.5.1 Characteristics:

- LEDs only allow current to pass in one direction from the positive (anode) to the negative (cathode). If connected the wrong way, they won't light up.
- They uses very little electricity, which makes them perfect for the battery-powered devices and energy-saving applications.
- LED turns on instantly and shine brightly, even at low voltage.
- Unlike regular bulb, LED don't have a filament, so they last for thousands of hours without burning out.
- You can get LEDs in different colours such as red, green, blue, yellow, and white each made with different materials.

3.2.5.2 Applications:

- Used in gadgets, chargers, and appliances to show the power or status.
- Found in bulbs, tube lights, and outdoor lights for energy efficient lightning.
- Used in digital clocks, calculators, TV displays, and advertising boards.
- Provide clear and long-lasting lighting for road safety and public information.
- Used to send signals in remotes and other communication system



Figure 3.5: - LED

3.2.6 9V BATTERY:

A 9V battery is a compact, rectangular battery commonly used to power small electronic devices. It's easy to recognize by its box type shape and two snap connectors on top one positive and another one is negative. This battery supplies a steady 9 volts of power, which is ideal for circuits and gadgets that need more voltage than smaller batteries can provide.

3.2.6.1 Characteristics:

- It provides a constant 9 volts of direct current (DC), which is useful for powering both the analog and digital devices.
- The 9V battery is small and lightweight, making it easy to carry and fit into compact places.
- It uses a snap-on connector for quick and secure connections, especially in electronics.
- Available in alkaline, lithium, or rechargeable versions, depending on power needs and reuse.

3.2.6.2 Applications:

- Used in homes to power the fire alarms.
- Toys and Gadgets: Powers remote-controlled toys and small devices
- Used in tools that check voltage and current.
- Perfect for Arduino projects and breadboard experiments when you need portable power.



Figure 3.6: - 9V Battery

CHAPTER 4

METHODOLOGY

This project is designed to control the brightness level of a light using claps detected by a sound sensor. The system uses an Arduino microcontroller, a sound sensor, and an LED. When the user claps, the sound sensor detects the sound, and based on the number of claps, the Arduino adjusts the brightness of the light. The sound is read from analog pin A0 which allows us to measure sound intensity more accurately.

4.1 Sound Detection Using Analog Pin A0:

We use a microphone based sound sensor module (LM393), which has an analog output pin connected to Arduinos analog pin A0. The sensor has four pins: VCC (5V), GND, D0(Digital Out), and A0 (Analog Out). In this project, we use the A0 pin, which gives varying voltages based on sound level. This voltage is read by Arduino analog pin A0 using analogRead(A0).

The Arduino continuously reads values from pin A0. When a clap sound detect, it causes a sudden spike in the analog value. By comparing the analog readings to a threshold value (greater than 600), the Arduino detects that the clap has detected. This method is more accurate and adjustable compared to using a digital pin.

4.2 Clap Counting and Timing Logic:

Each time the analog value crosses the threshold, a clap is counted. A clapCount variable is used to keep track of how many claps occur within a short time window for 300ms. A timer starts when the first clap is detected. If no new clap is detected within 300ms window, the total number of claps is finalized.

For example, if the user claps three times within two seconds, the Arduino will read three peaks from analog pin A0 and register a clapCount of 3.

4.3 Light Brightness Control:

After determining the number of claps, the Arduino adjusts the brightness of a light. The brightness is set using Pulse Width Modulation (PWM), which simulates analog

output using digital signals. The analogWrite() function is used on a PWM pin like pin 9.

The brightness levels are given as:

- 1 Clap → PWM = 64 (25% brightness)
- 2 Claps → PWM = 128 (50% brightness)
- 3 Claps → PWM = 192 (75% brightness)
- 4 Claps → PWM = 255 (100% brightness)
- 5 Claps → PWM = 0 (Light Off)

This value is sent to the LED . After setting the brightness levels, the clapCount is reset to 0, and the system starts listening again for new clap.

4.4 Circuit diagram:

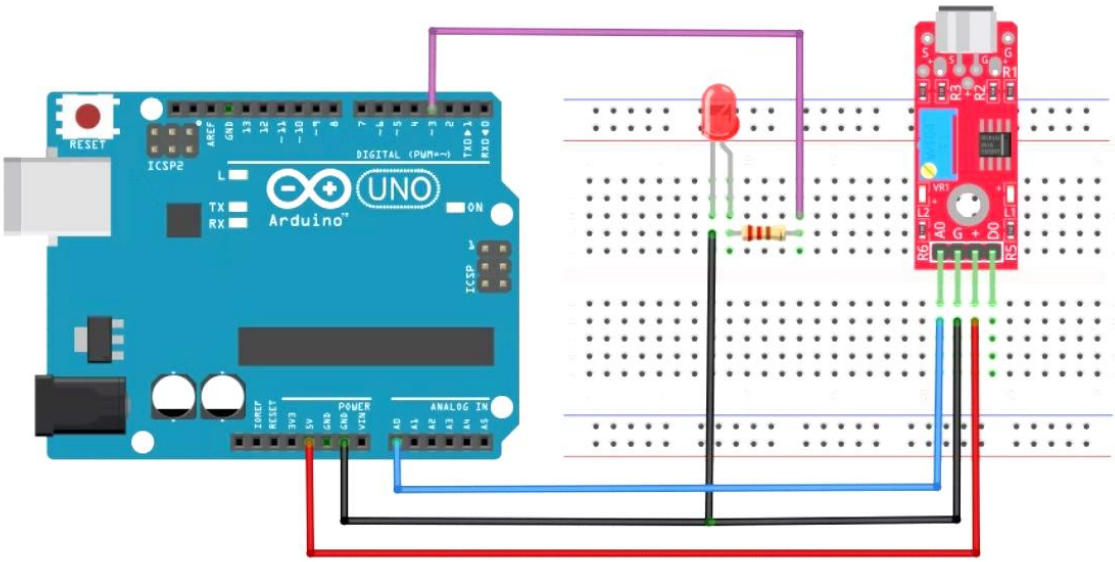


Figure 4.1: - Circuit Diagram of clap based light intensity control

4.5 Components:

Arduino UNO: A microcontroller.

LM393: Used to detect sound.

LED: Used for showing brightness level.

9V Battery: Provides Power supply to the Circuits.

Jumper wires: Used for connecting different part of the circuit.

4.6 Connections

4.6.1 Sound Sensor to Arduino:

VCC (+): Connected to the 5V pin of the Arduino.

GND(G): Connected to the GND of the Arduino.

Analog output (A0): connected to an Analog pin (A0) on the Arduino. This pin will detect the claps as high or low signals.

4.6.2 LED to Arduino:

Positive (anode): connected to a PWM digital pin 9 of the Arduino.

Negative (cathode): connected to GND of the Arduino.

4.6.3 Power Supply:

The Arduino can be powered via USB from a computer or a 9V battery using the Vin pin and GND.

4.7 Circuit Implementation:

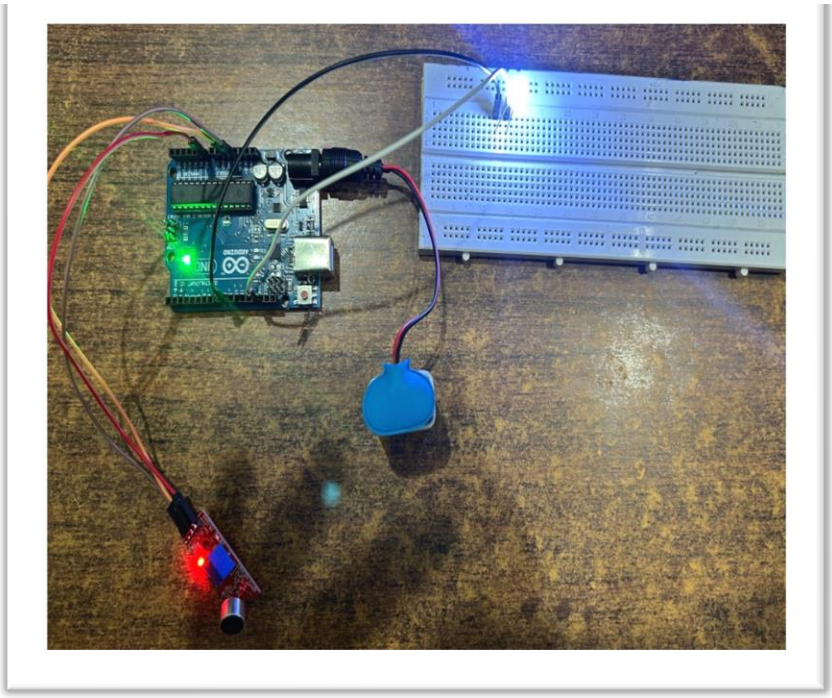


Figure 4.1: - Circuit Diagram of clap based light intensity control

APPENDIX

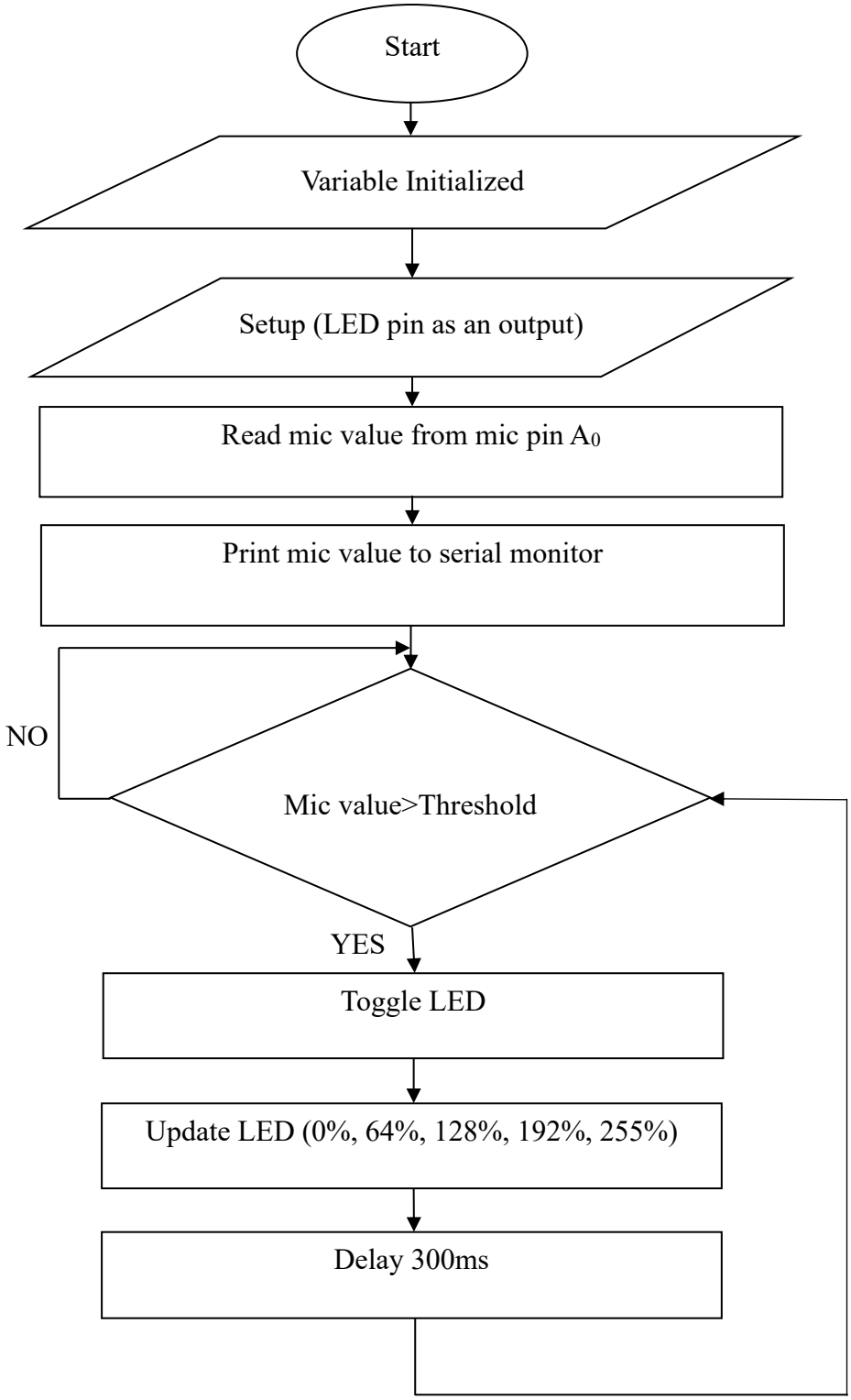
Code & Flowchart

1. Code:

Upload the following code on Arduino

```
int soundSensor =A0 ;
int ledPin= 9 ;
int clapThreshold = 600;
int brightnessLevels[] = {0, 64, 128, 192, 255};
int currentLevel = 0;
unsigned long lastClapTime = 0;
void setup() {
   pinMode(ledPin, OUTPUT);
   Serial.begin(9600);}
void loop() {
   int soundValue = analogRead(soundSensor);
   Serial.println(soundValue);
 if (soundValue > clapThreshold && millis() - lastClapTime > 300) {
       currentLevel = (currentLevel + 1) % 5;
       analogWrite(ledPin, brightnessLevels[currentLevel]);
       Serial.print("Brightness Set to: ");
       Serial.println(brightnessLevels[currentLevel]);
       lastClapTime = millis();
   }
}
```

2. Flowchart:


Start
Variable Initialized
Setup (LED pin as an output)
Read mic value from mic pin A0
Print mic value to serial monitor
NO
Mic value>Threshold
YES
Toggle LED
Update LED (0%, 64%, 128%, 192%, 255%)
Delay 300ms

CHAPTER-5

RESULT

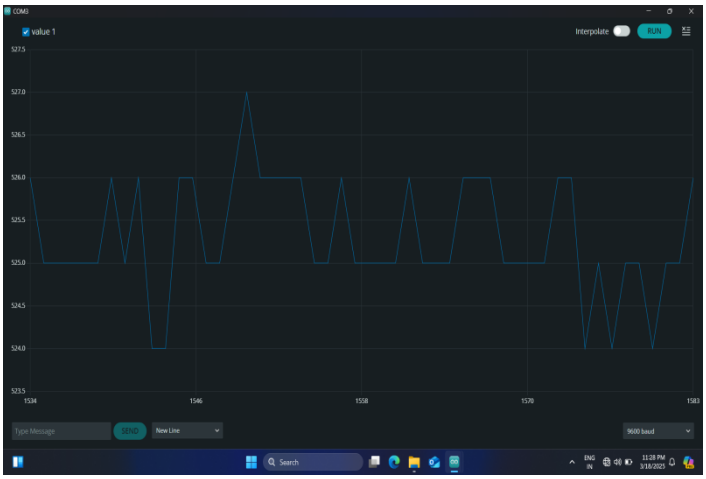


Figure 5.1: - Reading before clap

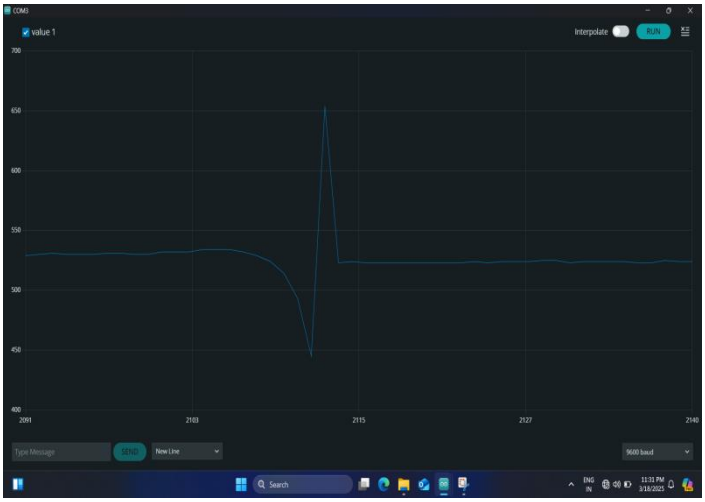


Figure 5.2: - Reading after clap

CHAPTER-6

FUTURE SCOPE & CONCLUSION

6.1 Future Scope:

6.1.1 Improved Sound Detection:

- Use better-quality microphones or sound sensors for more accurate clap detection.
- Implement noise filtering algorithms to avoid false triggers from background sounds.

6.1.2 Integration with Sensors:

- Combine with an LDR to respond only when it's dark.
- Add a motion sensor so the light activates only when someone is present.

6.1.3 Remote Access and IoT Integration:

- Use modules like ESP8266 or ESP32 to control the system via Wi-Fi or a mobile app.
- Add Bluetooth connectivity for wireless configuration or control.

6.1.4 Scalability:

- Expand the system to control multiple devices or rooms from one central unit.
- Create a modular design that can be deployed in smart homes or classrooms.

6.1.5 AI or ML Integration:

- Use machine learning models to better detect and adapt to different clapping styles or environmental noises.
- Train the system to personalize responses based on user habits.

6.2 Conclusion

This project shows that the very simple act of a clap can be used to control the lights using an Arduino. By adding a sound sensor and other simple components along with coding, we built a system that raises and lowers the light depending on the number of claps. This might seem insignificant but it is a step forward towards higher automation and control in the home. It not only provides an engaging and entertaining experience but also demonstrates the ease at which Arduino can be used to create something inspiring. With a few improvements like by adding wireless control, or IoT the idea can be turn into an advanced smart lighting system.

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